

Práctica 4 — VPN Client-to-Site: IPSec IKEv1 con L2TP

Estudiante: Ashley Fabian **Matrícula:** 2025-0773 **Institución:** Instituto Tecnológico Las Américas (ITLA)

Herramienta: GNS3 (CSR1000v, IOSvL2, VPCS) + VMware + Windows Client **Direccionamiento base:** 25.7.73.0/24

1. Introducción

Esta documentación describe la configuración de una **VPN Client-to-Site punto a multipunto**, utilizando **IPSec IKEv1 combinado con L2TP**, para permitir que un cliente Windows externo se conecte de forma segura a una LAN protegida a través de un túnel cifrado.

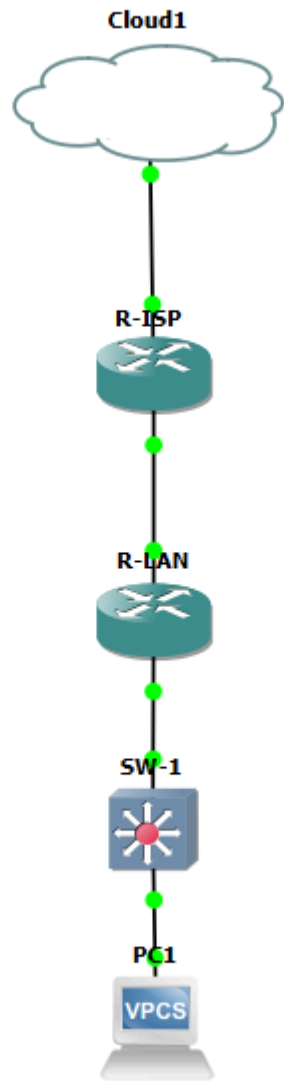
Esta es la novena y última topología de la Práctica 4, complementando las seis VPN site-to-site (IKEv1/IKEv2 en modalidad basada en políticas, VTI y GRE sobre IPSec) y las dos topologías DMVPN (Fase 2 y Fase 3).

2. Topología

```
[LAN-Host] --- [SW-LAN (IOSvL2)] --- [R-LAN (CSR1000v)] --- [R-ISP] ---  
[Cloud/NAT] --- [Windows Client VM]  
simulado)                                (VPN Server L2TP/IPSec)          (Internet
```

 **CAPTURA 1:** Vista completa de la topología en el workspace de GNS3, con todos los nodos visibles y las conexiones etiquetadas.

Ashley Fabian
2025-0773



2.1 Direcccionamiento

Segmento	Red	Notas
WAN R-LAN ↔ R-ISP	25.7.73.0/30	R-LAN: .2, R-ISP: .1
WAN R-ISP ↔ Cliente	25.7.73.4/30	R-ISP: .5, Cliente: .6
LAN interna (protegida)	25.7.73.32/27	R-LAN: .33 (gateway), Host: .34
Pool L2TP (IP virtual al cliente)	25.7.73.64/28	Asignado dinámicamente (.66–.78)

3. Configuración de los equipos

3.1 R-ISP

```
enable
configure terminal
hostname R-ISP

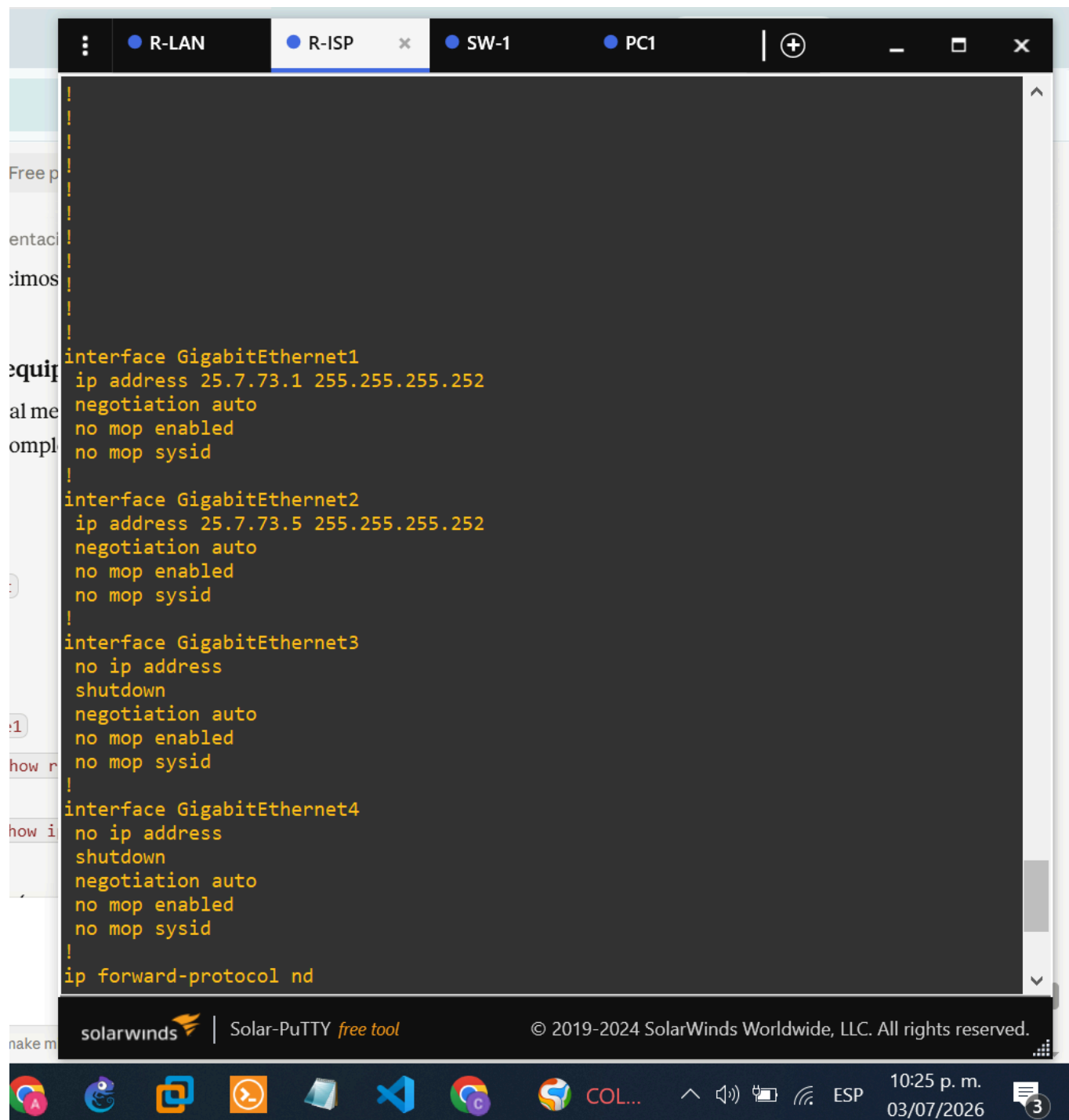
interface G1
 ip address 25.7.73.1 255.255.255.252
 no shutdown
 exit

interface G2
 ip address 25.7.73.5 255.255.255.252
 no shutdown
 exit

ip route 25.7.73.32 255.255.255.224 25.7.73.2

end
write memory
```

 **CAPTURA 2:** `show running-config` de R-ISP mostrando las interfaces G1 y G2 con sus IPs, y la ruta estática configurada.



The screenshot shows a SolarWinds Solar-PuTTY terminal window with a dark background and yellow text. The window title bar includes tabs for 'R-LAN', 'R-ISP', 'SW-1', and 'PC1'. The terminal displays the following configuration commands:

```
!
!
!
!
!
interface GigabitEthernet1
 ip address 25.7.73.1 255.255.255.252
 negotiation auto
 no mop enabled
 no mop sysid
!
interface GigabitEthernet2
 ip address 25.7.73.5 255.255.255.252
 negotiation auto
 no mop enabled
 no mop sysid
!
interface GigabitEthernet3
 no ip address
 shutdown
 negotiation auto
 no mop enabled
 no mop sysid
!
interface GigabitEthernet4
 no ip address
 shutdown
 negotiation auto
 no mop enabled
 no mop sysid
!
ip forward-protocol nd
```

The bottom of the window shows the SolarWinds logo, 'Solar-PuTTY free tool', and a copyright notice: '© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.' The Windows taskbar at the bottom shows the time as 10:25 p.m. on 03/07/2026.

3.2 R-LAN (CSR1000v) — Servidor VPN L2TP/IPSec IKEv1

```
enable
configure terminal
hostname R-LAN

interface G1
 ip address 25.7.73.2 255.255.255.252
 no shutdown
 exit

interface G2
 ip address 25.7.73.33 255.255.255.224
```

```
no shutdown
exit

ip route 0.0.0.0 0.0.0.0 25.7.73.1

! AAA local para autenticación PPP
aaa new-model
aaa authentication ppp L2TP-AUTH local
aaa authorization network L2TP-AUTHOR local

username vpnuser password 0 Vpn2025Fabian

! Política IKE Fase 1 (IKEv1)
! AES-256 / SHA / DH Group 14: coincide con la propuesta por defecto
! del cliente VPN nativo de Windows
crypto isakmp policy 5
  encr aes 256
  hash sha
  authentication pre-share
  group 14
  lifetime 86400
exit

crypto isakmp key Preshared2025Key address 25.7.73.6

! Transform-set IPSec en modo transporte (obligatorio para L2TP)
crypto ipsec transform-set L2TP-SET esp-aes esp-sha-hmac
  mode transport
exit

! ACL de tráfico interesante: UDP 1701 en origen Y destino,
! que es como el cliente L2TP nativo de Windows negocia el proxy-ID
access-list 100 permit udp any eq 1701 any eq 1701

! Crypto map dinámico (el peer no tiene IP fija garantizada)
crypto dynamic-map L2TP-DYNMAP 10
  set transform-set L2TP-SET
  match address 100
exit

crypto map L2TP-MAP 10 ipsec-isakmp dynamic L2TP-DYNMAP

interface G1
  crypto map L2TP-MAP
exit

! VPDN (servidor L2TP)
vpdn enable

vpdn-group L2TP-GROUP
  accept-dialin
  protocol l2tp
  virtual-template 1
exit
```

```
no l2tp tunnel authentication
exit

! Virtual-Template: asigna IP al cliente y autentica PPP
ip local pool L2TP-POOL 25.7.73.66 25.7.73.78

interface Virtual-Template1
 ip unnumbered G2
 peer default ip address pool L2TP-POOL
 ppp authentication ms-chap-v2 L2TP-AUTH
 exit

end
write memory
```

 **CAPTURA 3:** `show running-config` completo de R-LAN, visible desde `crypto isakmp policy 5` hasta `interface Virtual-Template1`.

7 / 20

```
multilink bundle-name authenticated
vpdn enable
!
vpdn-group L2TP-GROUP
! Default L2TP VPDN group
accept-dialin
protocol l2tp
virtual-template 1
no l2tp tunnel authentication
!
!
!
!
!
!
!
!
!
!
license udi pid CSR1000V sn 9DB961S3K3J
no license smart enable
diagnostic bootup level minimal
!
spanning-tree extend system-id
!
!
!
username vpnuser password 0 Vpn2025Fabian
!
redundancy
!
!
--More--
```

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```
!
crypto isakmp policy 5
  encr aes 256
  authentication pre-share
  group 14
!
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
crypto isakmp key Preshared2025Key address 25.7.73.6
!
crypto ipsec transform-set L2TP-SET esp-aes esp-sha-hmac
mode transport
!
crypto dynamic-map L2TP-DYNMAP 10
  set transform-set L2TP-SET
  match address 100
!
crypto map L2TP-MAP 10 ipsec-isakmp dynamic L2TP-DYNMAP
!
!
interface GigabitEthernet1
  ip address 25.7.73.2 255.255.255.252
  negotiation auto
  no mop enabled
--More-- █
```

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10:27 p. m.
03/07/2026

The screenshot shows a SolarWinds Solar-PuTTY terminal window with a dark background and yellow text. The window title bar includes tabs for 'R-LAN', 'R-ISP', 'SW-1', and 'PC1'. The configuration text is as follows:

```
crypto map L2TP-MAP 10 ipsec-isakmp dynamic L2TP-DYNMAP
!
!
!
!
interface GigabitEthernet1
ip address 25.7.73.2 255.255.255.252
negotiation auto
no mop enabled
no mop sysid
crypto map L2TP-MAP
!
interface GigabitEthernet2
ip address 25.7.73.33 255.255.255.224
negotiation auto
no mop enabled
no mop sysid
!
interface GigabitEthernet3
no ip address
shutdown
negotiation auto
no mop enabled
no mop sysid
!
interface GigabitEthernet4
no ip address
shutdown
negotiation auto
no mop enabled
no mop sysid
!
--More--
```

The bottom of the window shows the SolarWinds logo, the text 'Solar-PuTTY free tool', and the copyright notice '© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.' The Windows taskbar at the bottom displays various application icons, the system clock showing 10:28 p.m. on 03/07/2026, and the temperature 27°C.



The screenshot shows a SolarWinds Solar-PuTTY terminal window with a dark background and yellow text. The window title bar includes tabs for 'R-LAN', 'R-ISP', 'SW-1', and 'PC1'. The configuration text is as follows:

```
!
ip local pool L2TP-POOL 25.7.73.66 25.7.73.78
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip route 0.0.0.0 0.0.0.0 25.7.73.1
!
!
access-list 100 permit udp any eq 1701 any eq 1701
!
!
!
!
!
control-plane
!
!
!
!
!
!
line con 0
  stopbits 1
line vty 0 4
!
wsma agent exec
!
wsma agent config
!
wsma agent filesys
!
wsma agent notify
!
!
end
R-LAN#
```

The bottom of the window shows the SolarWinds logo, the text 'Solar-PuTTY free tool', and a copyright notice: '© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.' Below this is a Windows taskbar with various icons, including a clock showing 10:28 p.m. on 03/07/2026.

3.3 SW-LAN (IOSvL2)

```
enable
configure terminal
hostname SW-LAN

vlan 10
  name LAN_PROTEGIDA
  exit

interface range Gi0/0 - 1
```

```
switchport mode access
switchport access vlan 10
no shutdown
exit

end
write memory
```

 **CAPTURA 4:** `show vlan brief` en SW-LAN mostrando la VLAN 10 con los puertos Gi0/0 y Gi0/1 asignados.

```
SW-LAN#sh vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/2, Gi0/3, Gi1/0, Gi1/1 Gi1/2, Gi1/3, Gi2/0, Gi2/1 Gi2/2, Gi2/3, Gi3/0, Gi3/1 Gi3/2, Gi3/3
10	LAN_PROTEGIDA	active	Gi0/0, Gi0/1
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
SW-LAN#
```

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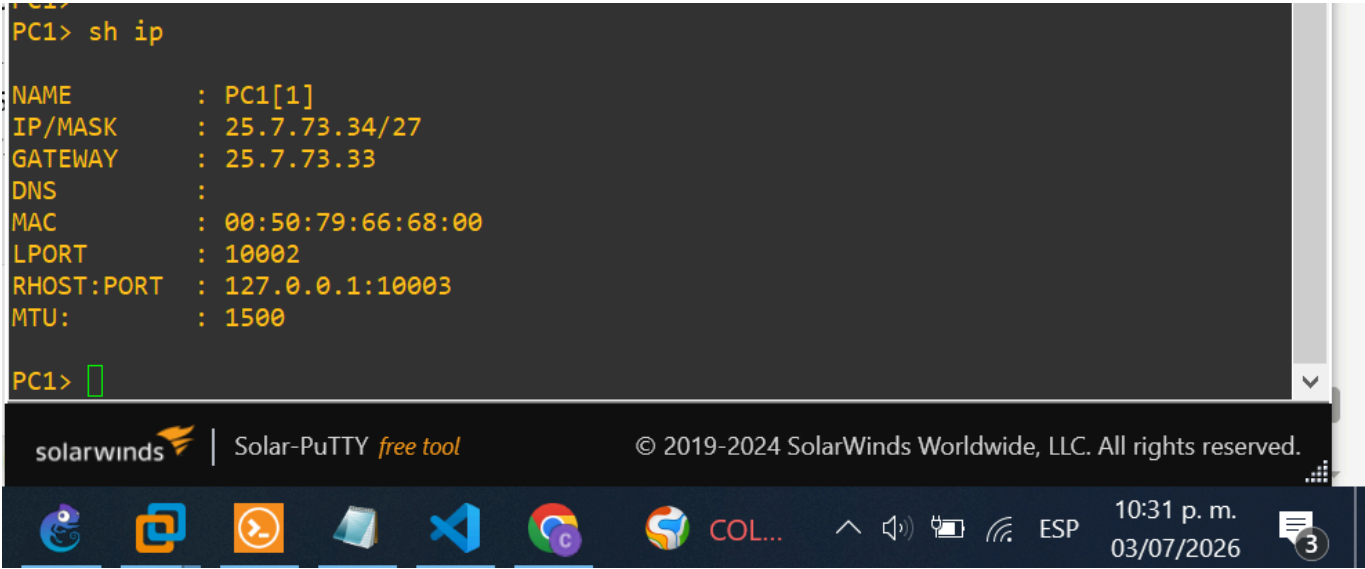
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3.4 LAN-Host (VPCS)

```
ip 25.7.73.34 255.255.255.224 25.7.73.33
save
```

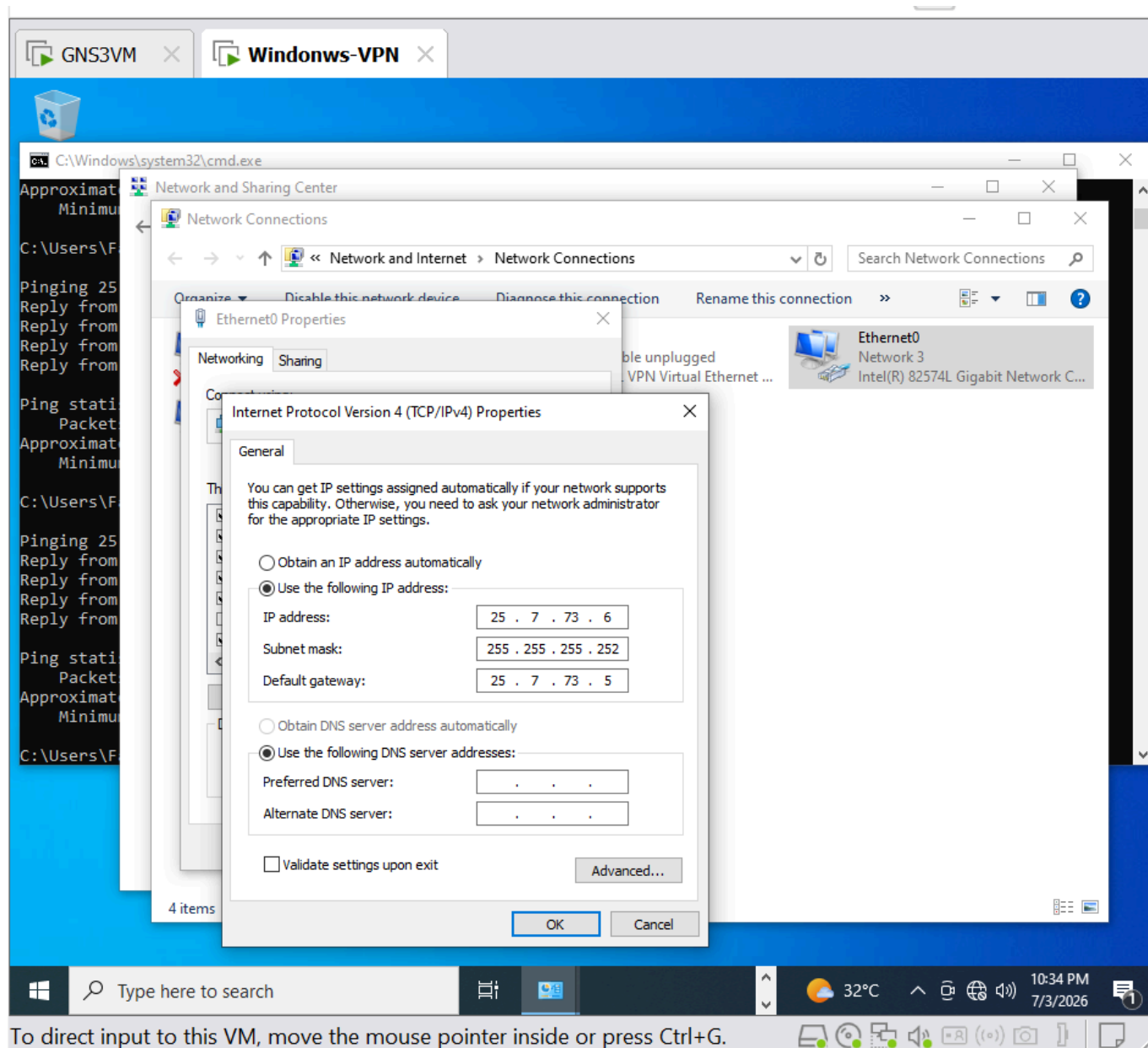
 **CAPTURA 5:** Salida de `show ip` en LAN-Host confirmando IP y gateway.



3.5 Windows Client VM — Adaptador de red (segmento público)

Parámetro	Valor
IP	25.7.73.6
Máscara	255.255.255.252
Puerta de enlace	25.7.73.5

 **CAPTURA 6:** Ventana de propiedades de "Protocolo de Internet versión 4 (TCP/IPv4)" con estos valores configurados.

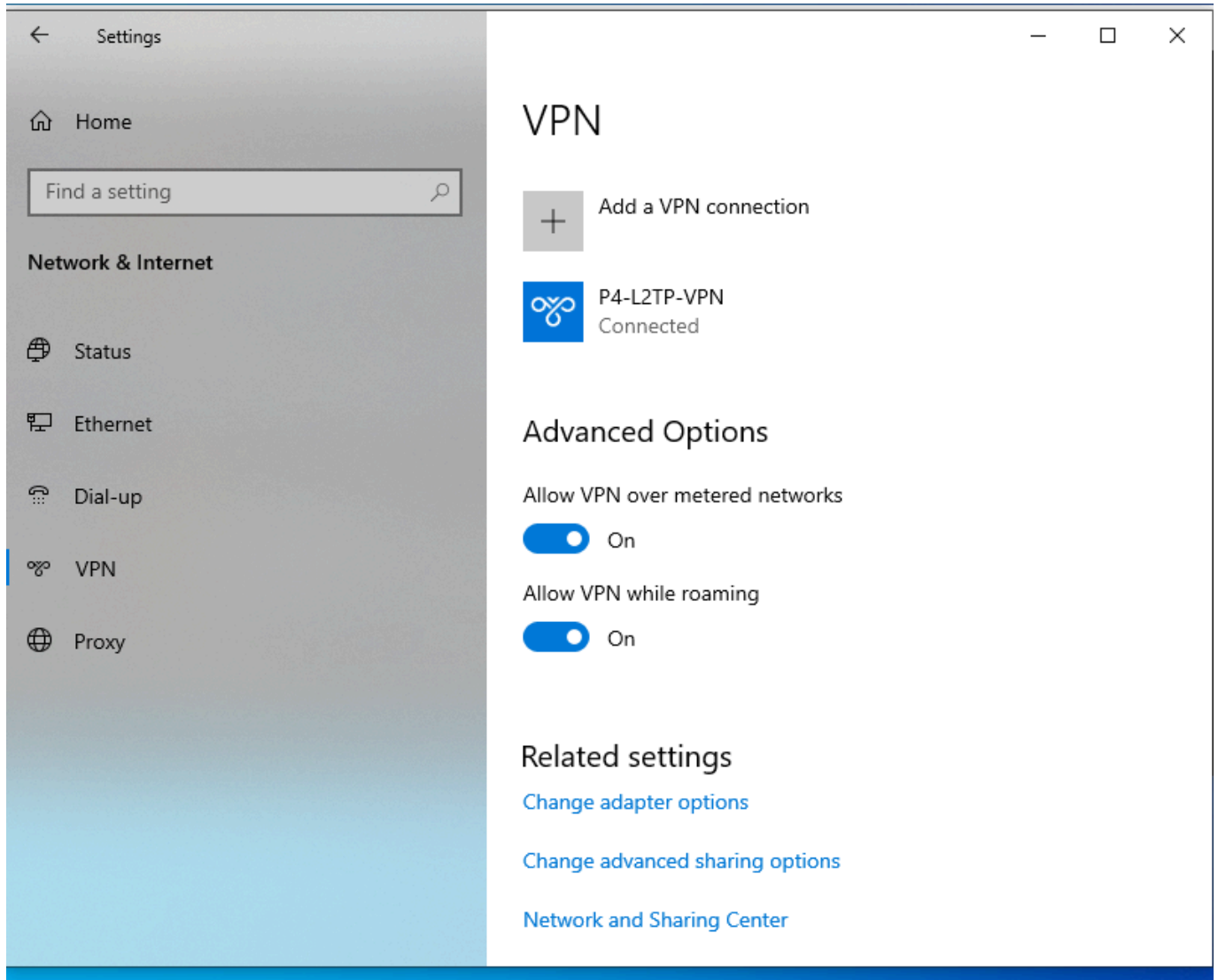


3.6 Windows Client VM — Conexión VPN

Configuración → Red e Internet → VPN → Agregar una conexión VPN

Parámetro	Valor
Proveedor de VPN	Windows (integrado)
Nombre de conexión	P4-L2TP-VPN
Servidor	25.7.73.2
Tipo de VPN	L2TP/IPSec con clave precompartida
Clave precompartida	Preshared2025Key
Usuario	vpnuser
Contraseña	Vpn2025Fabian

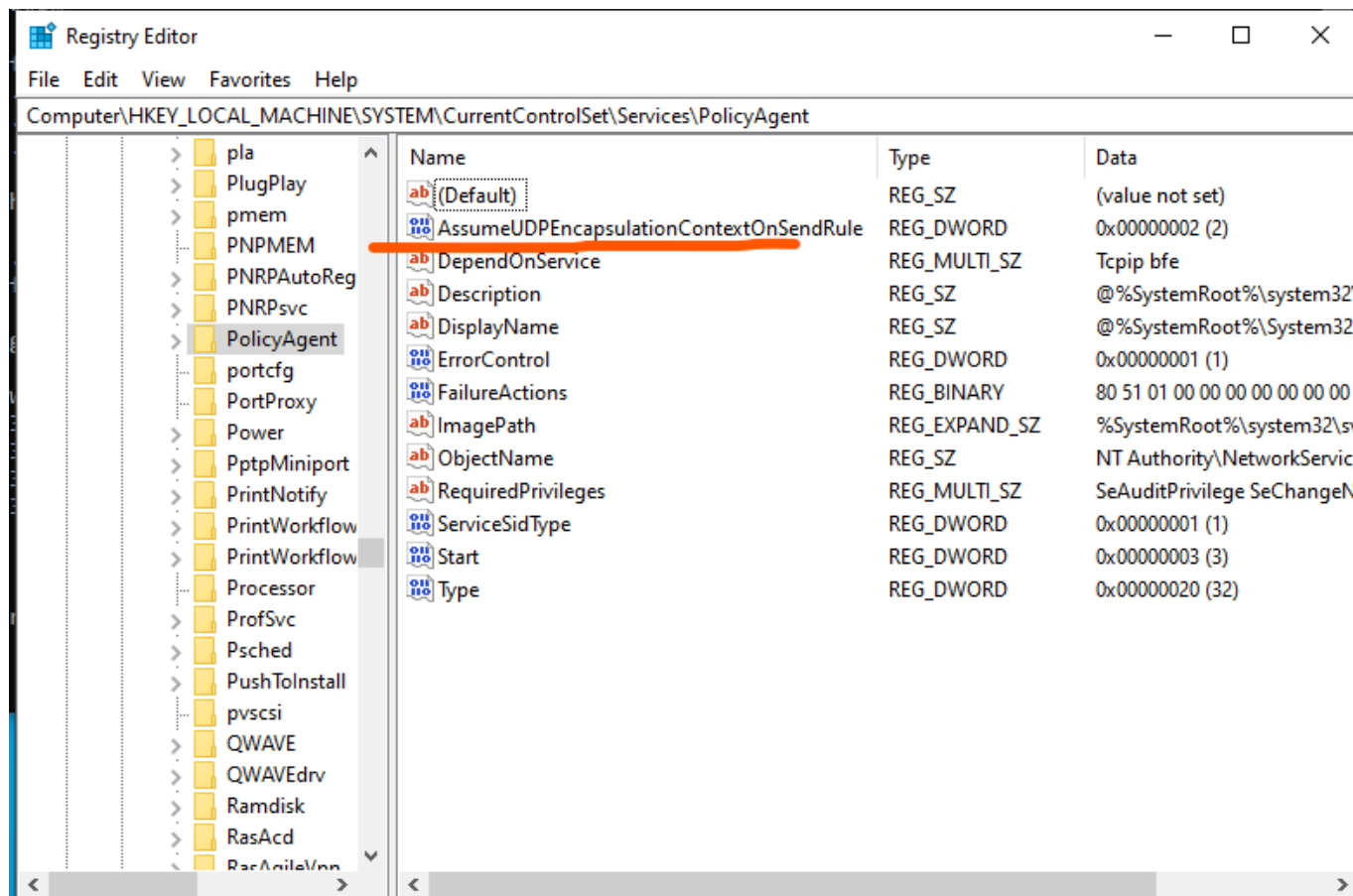
 **CAPTURA 7:** Pantalla de configuración de la conexión VPN en Windows con todos los campos anteriores visibles.



Ajuste de registro (requerido porque el servidor VPN no es Windows):

```
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\PolicyAgent
Nombre: AssumeUDPEncapsulationContextOnSendRule (DWORD 32 bits)
Valor: 2
```

 **CAPTURA 8:** Ventana de `regedit` mostrando la clave `AssumeUDPEncapsulationContextOnSendRule` creada con valor `2`.

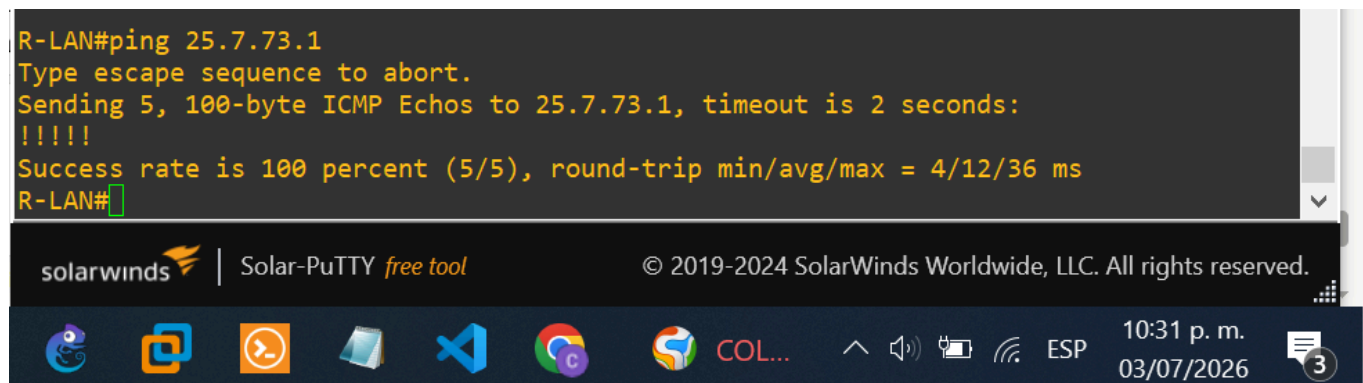


4. Verificación de conectividad básica (pre-VPN)

Antes de establecer el túnel, se confirmó conectividad IP de extremo a extremo sin cifrado:

```
R-LAN# ping 25.7.73.1
PC1> ping 25.7.73.33
```

CAPTURA 9: Ping exitoso de R-LAN hacia R-ISP (25.7.73.1).



CAPTURA 10: Ping exitoso de LAN-Host hacia su gateway (25.7.73.33).


```
PC1> ping 25.7.73.33
84 bytes from 25.7.73.33 icmp_seq=1 ttl=255 time=26.112 ms
84 bytes from 25.7.73.33 icmp_seq=2 ttl=255 time=17.198 ms
84 bytes from 25.7.73.33 icmp_seq=3 ttl=255 time=23.605 ms
84 bytes from 25.7.73.33 icmp_seq=4 ttl=255 time=33.570 ms
84 bytes from 25.7.73.33 icmp_seq=5 ttl=255 time=33.888 ms

PC1> █
```

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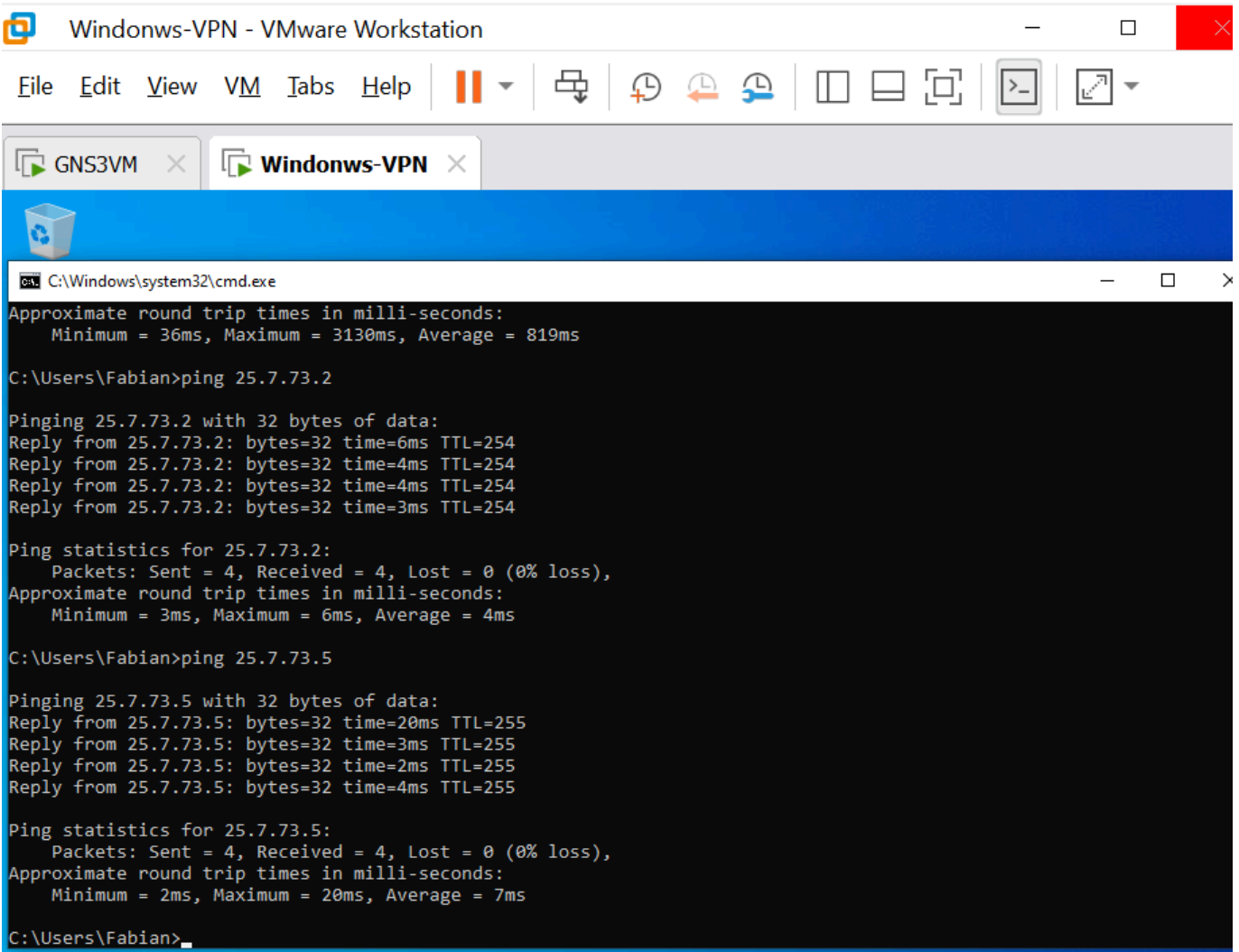
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10:32 p. m.
03/07/2026

Desde Windows:

```
ping 25.7.73.5
ping 25.7.73.2
```

 **CAPTURA 11:** Ambos pings exitosos desde la terminal (cmd) de Windows.

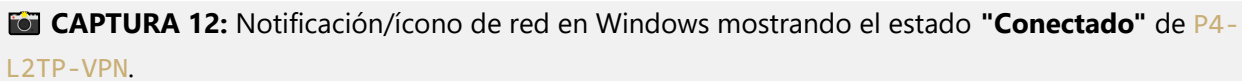


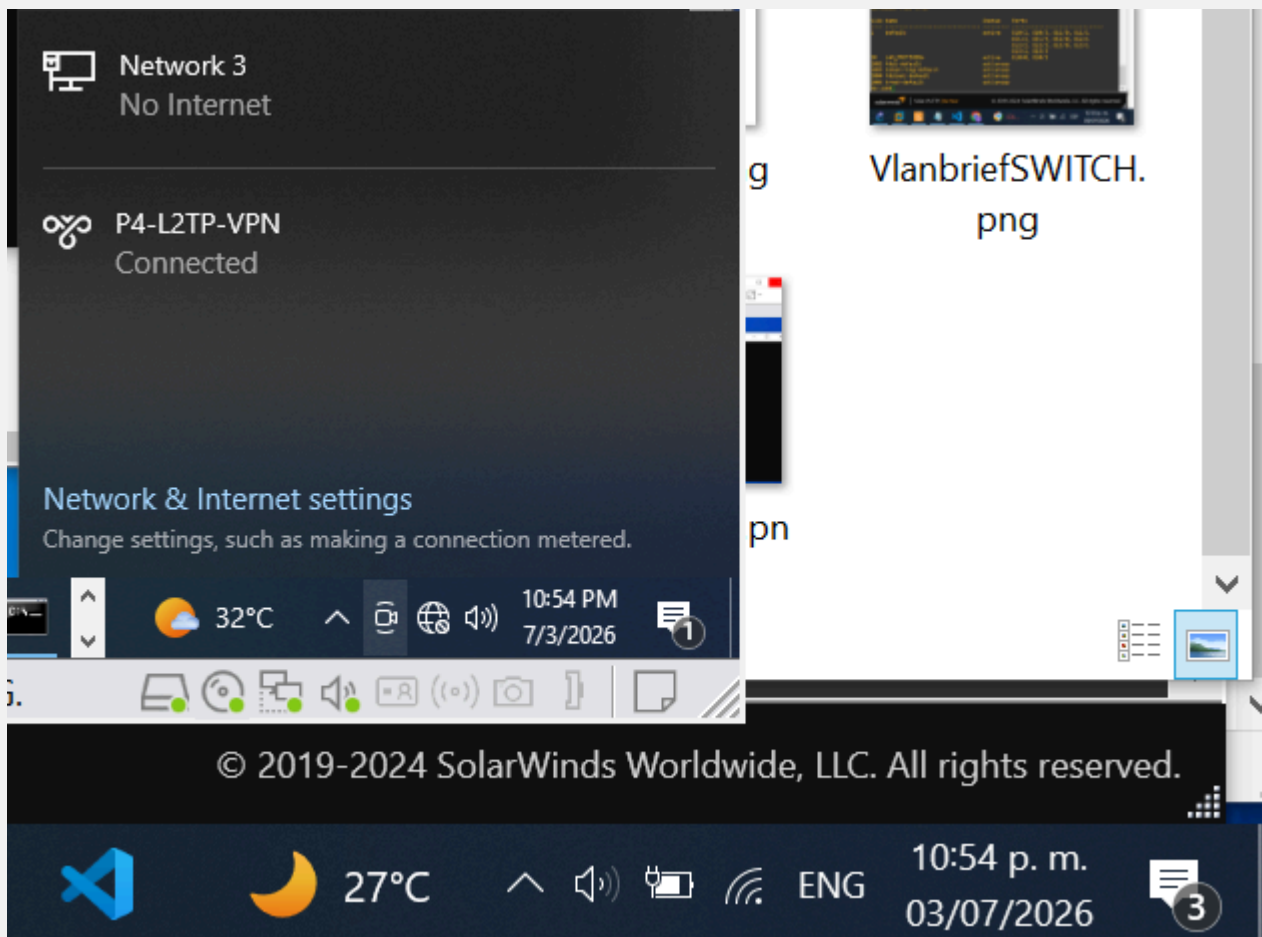
```
Windows-VPN - VMware Workstation
File Edit View VM Tabs Help
GNS3VM Windonws-VPN
C:\Windows\system32\cmd.exe
Approximate round trip times in milli-seconds:
  Minimum = 36ms, Maximum = 3130ms, Average = 819ms
C:\Users\Fabian>ping 25.7.73.2
Pinging 25.7.73.2 with 32 bytes of data:
Reply from 25.7.73.2: bytes=32 time=6ms TTL=254
Reply from 25.7.73.2: bytes=32 time=4ms TTL=254
Reply from 25.7.73.2: bytes=32 time=4ms TTL=254
Reply from 25.7.73.2: bytes=32 time=3ms TTL=254
Ping statistics for 25.7.73.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 6ms, Average = 4ms
C:\Users\Fabian>ping 25.7.73.5
Pinging 25.7.73.5 with 32 bytes of data:
Reply from 25.7.73.5: bytes=32 time=20ms TTL=255
Reply from 25.7.73.5: bytes=32 time=3ms TTL=255
Reply from 25.7.73.5: bytes=32 time=2ms TTL=255
Reply from 25.7.73.5: bytes=32 time=4ms TTL=255
Ping statistics for 25.7.73.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 20ms, Average = 7ms
C:\Users\Fabian>
```

5. Establecimiento del túnel VPN

Con **debug crypto isakmp** activo en R-LAN, se inició la conexión desde Windows. La negociación pasó por:

1. **Fase 1 (ISAKMP):** autenticación exitosa con **SA has been authenticated with 25.7.73.6**.
2. **Fase 2 (IPSec Quick Mode):** SA instalada con el transform-set **esp-aes esp-sha-hmac** en modo transporte.
3. **Subida de la interfaz Virtual-Access:** **%LINEPROTO-5-UPDOWN: Line protocol on Interface Virtual-Access2, changed state to up.**

 **CAPTURA 12:** Notificación/ícono de red en Windows mostrando el estado "**Conectado**" de **P4-L2TP-VPN**.



6. Verificación final

En R-LAN:

```
show vpdn session
show users
show ip interface brief | include Virtual-Access
```

 **CAPTURA 13:** Salida de **show vpdn session** mostrando la sesión activa de **vpnuser**.

```
R-LAN#show vpdn session

L2TP Session Information Total tunnels 1 sessions 1

LocID      RemID      TunID      Username, Intf/      State  Last Chg Uniq ID
          Vcid, Circuit
32377      1          28931      vpnuser, Vi2.1      est    00:57:10 1
R-LAN#
```

CAPTURA 14: Salida de `show users` mostrando la interfaz `Vi2.1`, usuario `vpnuser` e IP `25.7.73.66`.

```
R-LAN#show users

Line      User      Host(s)      Idle      Location
*  0 con 0      idle      00:00:00

Interface  User      Mode      Idle      Peer Address
Vi2.1      vpnuser   PPPoVPDN   -         25.7.73.66
R-LAN#
```

CAPTURA 15: Salida de `show ip interface brief | include Virtual-Access` con `Virtual-Access2` en estado `up/up`.

```
R-LAN#show ip interface brief | include Virtual-Access
Virtual-Access1      unassigned      YES unset      down      down
Virtual-Access2      unassigned      YES unset      up        up
Virtual-Access2.1    25.7.73.33      YES unset      up        up
R-LAN#
```

En Windows:

```
ipconfig
ping 25.7.73.34
```

CAPTURA 16: Salida de `ipconfig` mostrando la IP virtual recibida (`25.7.73.66`).

```
C:\Windows\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::e9cd:f37a:f2bc:7aa%9
    IPv4 Address. . . . . : 25.7.73.6
    Subnet Mask . . . . . : 255.255.255.252
    Default Gateway . . . . . : 25.7.73.5

PPP adapter P4-L2TP-VPN:

    Connection-specific DNS Suffix  . :
    IPv4 Address. . . . . : 25.7.73.66
    Subnet Mask . . . . . : 255.255.255.255
    Default Gateway . . . . . : 0.0.0.0

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\Fabian>
```

 **CAPTURA 17:** Ping exitoso desde Windows hacia 25.7.73.34 (LAN-Host), confirmando acceso completo a la red protegida a través del túnel.

```
C:\Users\Fabian>ping 25.7.73.34

Pinging 25.7.73.34 with 32 bytes of data:
Reply from 25.7.73.34: bytes=32 time=3097ms TTL=63
Reply from 25.7.73.34: bytes=32 time=16ms TTL=63
Reply from 25.7.73.34: bytes=32 time=22ms TTL=63
Reply from 25.7.73.34: bytes=32 time=29ms TTL=63

Ping statistics for 25.7.73.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 3097ms, Average = 791ms

C:\Users\Fabian>
```

7. Conclusión

Se estableció exitosamente una VPN Client-to-Site basada en IPSec IKEv1 con L2TP, permitiendo que un cliente Windows externo accediera a la LAN protegida detrás de R-LAN. La configuración incluyó autenticación AAA local, una política IKE alineada con las propuestas del cliente nativo de Windows, un transform-set en modo transporte, y un pool de direcciones asignado dinámicamente vía PPP/L2TP. Con esta topología se completan las nueve VPN requeridas en la Práctica 4.